

Course Number	0418-331	Course Title	Computer Networks
Credit Hours	4 (3:3:4)	Course Coordinator	Dr. Mohammed Al-Mutawaa
Compulsory	Yes	Prerequisite	0418-222 or 0418-223

Textbook

<i>Title:</i>	<i>Computer Networking: A Top-Down Approach</i>
<i>Author:</i>	<i>James F. Kurose, Keith W. Ross</i>
<i>Publisher:</i>	<i>Pearson, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Computer Networks Overview: Network concepts, Protocol layers and Internet protocol stack, Throughput, delay and loss.	2
2	Application Layer: Web (HTTP, cookies, proxies), File transfer (FTP), Email (SMTP, POP, and IMAP), Name Resolution (DNS), Peer-to-Peer.	6
3	Transport Layer: Transport layer services, Multiplexing and Demultiplexing, Connectionless Transport (UDP), Connection-Oriented Transport (TCP), Reliable Data Transfer, Congestion Control.	6
4	Network Layer: Virtual and Datagram networks, IP Addressing, IP Routing, Network Address Translation, Broadcast and Multicast	6
5	Data Link Layer: Access Networks, Multiple Access Protocols, Channel Partitioning, Error Detection and Correction, Local Area Networks (Ethernet, ARP), Data Center Networking.	6
6	Wireless and Mobility: Wireless links, CDMA, WiFi, Mobility Management	3
7	Network Security: Network Security, Cryptography, Message Integrity, Authentication, Securing Email, Securing TCP, Operational Security	5
8	Parallel and Distributed Computing: Intro to Parallel and Distributed Systems, Distributed Systems Architectures and Models, Cloud Computing.	6
9	Exams	2
	Total	42

Laboratory work

	Laboratory	Weeks
1	Introduction to Computer Networks	1
2	Web and HTTP Protocol	2
3	Network Utilities and Tools	3
4	DNS	4
5	Socket Programming	5
6	UDP	6
7	TCP	7
8	IP and NAT	8
9	IP Routing	9
10	Ethernet	10
11	ARP	11
12	Network Topology and Design	12
13	Network Management	13
14	Security	14

Grading

15-20%	Midterm 1
15-20%	Midterm 2
15-20%	Lab
0-15%	Hmk/Quiz
40%	Final

Course Outcomes

Students should be able to:

1. Understand computer network concepts and different network architectures.
2. Understand common network application protocols.
3. Understand the functionality and responsibility of the lower four internet protocol stack layers and the various protocols related to them.
4. Develop simple network applications.
5. Analyze, evaluate and design different network topologies.
6. Understand the fundamentals of network security, threats and solutions, and the various protocol related to them.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	H	M				
2	H	M				
3	H	M				
4	M	H				M
5	H	H				M
6	H	M				